81. (c) 200 – 300° C — Kerosene fraction condenses in the boiling range of about 200 – 300° C.

82. (a) — Incorrect, because alkanes can be both straight-chain and branched; they are not all straight-chain compounds.

83. (d) Anti-knocking properties of fuel increases.

84. (a) $\underset{\text{Hexane Liquid}}{C_6H_{14}} \xrightarrow{\Delta} \underset{\text{Butane}}{C_4H_{10}} + \underset{\text{Ethene Gas}}{C_2H_2}$

85. **(b) Le Bel and Van't Hoff** — They first proposed the tetrahedral arrangement of four valencies around carbon (1874).

86. **(a) Frankland's reaction** — Alkyl halides react with zinc to form alkanes (coupling reaction).

87. (a) Compounds having active hydrogen $(ROH, H_2O, R-NH_2)$ can form alkane when treated with Grignard's reagent

$$CH_3CH_2OH + CH_3MgBr \rightarrow CH_4 + Mg < \begin{matrix} Br \\ OCH_2CH_3 \end{matrix}$$

88. (d) n-Pentane — n-Pentane is a liquid at STP; ethane, propane and n-butane are gases.

89. (a) — Not true, because alkanes are non-polar and insoluble in water.

90. (a) Synthetic petrol — Fischer – Tropsch process converts CO and H_2 (syngas) into synthetic hydrocarbons (fuel).

91. (a) It is not possible to prepare CH_4 by wurtz reaction.

92. (b) Octane number is the percentage by volume of iso-octane in the mixture of iso-octane and *n*-heptane which has the same antiknocking properties as the fuel under examination.

Given fuel (25% *n*-heptane +75% iso-octane) Hence, octane number = 75 (because iso octane is 75%)

93. (c) $CH_3-CH_2-CH_2-Br \xrightarrow[\text{Dehydro halogenati on}]{C_2H_5ONa} CH_3-CH=CH_2+HBr$

94. (a) % of hydrogen = $\frac{\text{Mass of hydrogen}}{\text{Mass of compound}} \times 100$

$$CH_4 = \frac{4}{16} \times 100 = 25\%.$$

95. (c) Molecular mass can be obtained by the victor mayer process

$$\text{Molecular mass} = \frac{\text{Weight}}{V\,ml.} \times 22400$$

$$= \frac{11}{5600} \times 22400 = 44$$

96. $CH_3 - C(CH_3)_2 - CH_2 - CH(CH_3) - CH_3$ — Iso-octane is 2,2,4-trimethylpentane, the reference compound assigned an octane number of 100.

97. (a)

$$\underset{\text{2,3-dibromo-3-methyl pentane}}{CH_3 - \underset{Br}{\underset{|}{CH}} - \underset{Br}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}} - CH_2 - CH_3} + Zn \rightarrow CH_3 - CH = \overset{CH_3}{\overset{|}{C}} - CH_2 - CH_3 + ZnBr_2$$

HI + Red P (Reduction)

$$\downarrow$$

$$\underset{\text{3-Methyl pentane}}{CH_3 - CH_2 - \underset{CH_3}{\underset{|}{CH}} - CH_2 - CH_3}$$

98. (d) Gasoline, kerosene oil, diesel

99. (a) $CH_3 - CH_2 - COONa \xrightarrow[\Delta]{\text{Soda lime}} CH_3 - CH_3$

100. (d) Gasoline or petrol composition $C_7 - C_{12}$.